

Document S2

Note S1. Biome- and vegetation-type-specific model evaluation

Methods. To assess whether vegetation-specific calibration could improve retrieval performance relative to a single global model (henceforth *generic* model), we repeated model calibration for groups defined by combinations of biome and vegetation type. Rather than defining groups by biome or land cover alone, we grouped biome $\times$ land-cover combinations representing similar vegetation structure and environmental conditions, while ensuring sufficient data for model calibration and evaluation. Biome assignments followed the RESOLVE ecoregion biome labels. Land-cover classes were derived from the NLCD labels associated with each GROUNDED-EO reference measurement (RM). We simplified the original NLCD classification into six vegetation classes: cropland, deciduous/mixed forest, evergreen forest, grassland, shrubland, and wetland. We then cross-tabulated these vegetation classes with biome assignments and combined reference measurements into eight groups (Table S10). Combinations that could not be assigned to a sufficiently represented group were excluded, removing 342 RMs. The resulting grouped dataset contained 10,696 RMs across 44 ecoregions.

We repeated the model calibration described in the main Methods independently for each biome-vegetation group and for a generic model using all RMs assigned to the retained groups. The same procedure and parameter search space were used for all models and traits. All models used the S2BIOPHYS hybrid PROSAIL inversion with a multilayer perceptron. Predictors comprised ten Sentinel-2 reflectance bands and three observation-geometry angles. To reduce computational costs, we used a reduced optimization budget of 1,500 trials per model, comprising 600 random startup trials followed by 900 trials using the tree-structured Parzen estimator (TPE). The generic model was calibrated separately for this experiment using the same reduced optimization budget.

Instead of the five-fold ecoregion-blocked cross-validation used for the main S2BIOPHYS model, we used three folds to accommodate the smaller number of ecoregions within individual biome-vegetation groups. All RMs from the same ecoregion were retained within the same fold.

For the primary comparison, we used stacked out-of-sample (OOS) predictions, with each RM predicted only when its ecoregion was held out. Vegetation-specific and generic models were evaluated on identical RMs within each group using RMSE and R^2 . The group-specific R^2 used the mean of each validation group. The global-reference R_g^2 used the mean across all groups. We defined $\Delta\text{RMSE} = \text{RMSE}_{\text{specific}} - \text{RMSE}_{\text{generic}}$, with negative values indicating better vegetation-specific performance.

We quantified uncertainty in ΔRMSE by paired cluster bootstrap resampling of the existing OOS predictions. We sampled ecoregions with replacement and retained all RMs from each sampled ecoregion. The same resampled clusters were used for both models. We used 5,000 replicates and percentile-based 95% confidence intervals. Intervals were reported only for validation groups with at least 10 ecoregions. Groups with fewer ecoregions retained point estimates because their spatial replication was insufficient for reliable bootstrap intervals.

As a sensitivity analysis, we also used a paired bootstrap of individual RMs. We sampled RMs with replacement within each validation group and across the pooled dataset. The same sampled RMs were used for both models. We used 5,000 replicates and percentile-based 95% confidence intervals. These intervals treat RMs as independent and do not account for spatial dependence within ecoregions.

Results. Overall, the generic model performed better when pooling RMs across all validation groups, with lower pooled RMSE for all three traits (Figure S10; Table S11). Vegetation-specific calibration provided improvements primarily in well-represented groups. For LAIe, specific models reduced RMSE in the three largest groups, whereas performance was generally less consistent in groups with fewer RMs. This indicates that vegetation-specific calibration can improve retrievals when sufficient reference data are available, but that pooling data across

vegetation types provides more robust performance overall. Bootstrap intervals were therefore estimated for the three temperate groups and for the pooled comparison. Groups with four or five ecoregions are shown without intervals because spatial replication was insufficient. All reported 95% confidence intervals included zero. The individual-RM bootstrap produced intervals for every group (Figure S11). These intervals are descriptive because they do not preserve the ecoregion-level dependence structure.

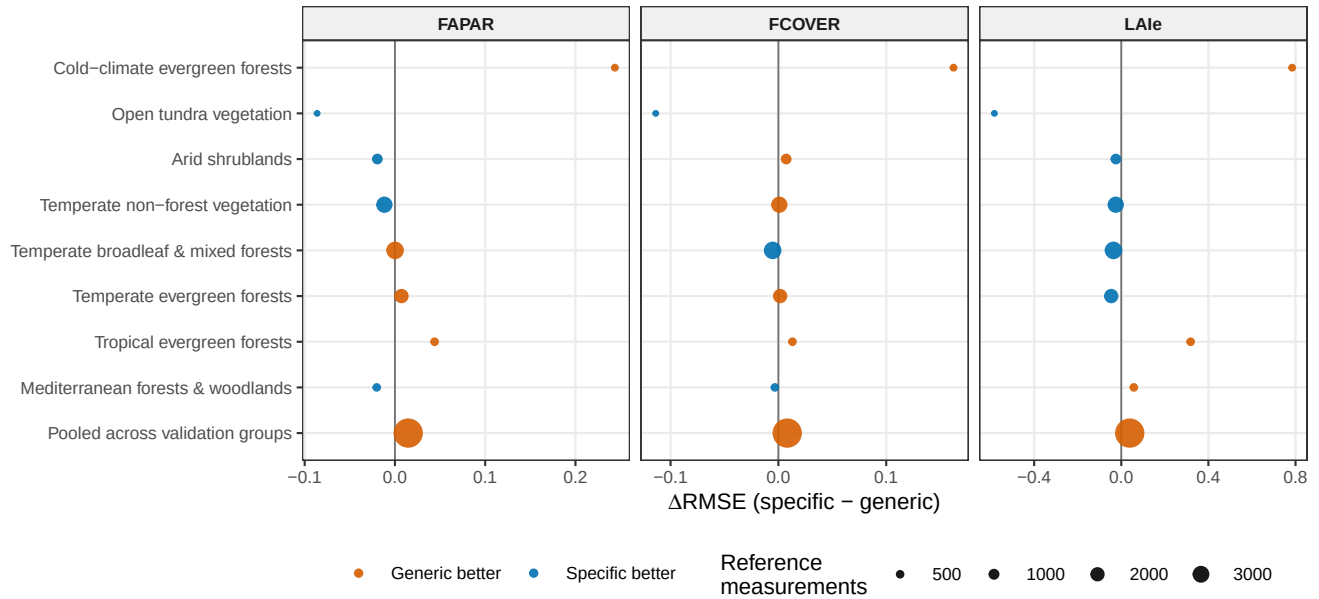


Figure S10: Change in out-of-sample RMSE for biome-vegetation-specific models relative to the generic model. Horizontal lines are paired ecoregion-bootstrap 95% confidence intervals and are shown only for groups with at least 10 ecoregions. Negative values indicate lower error for the specific model. Point size represents the number of reference measurements.

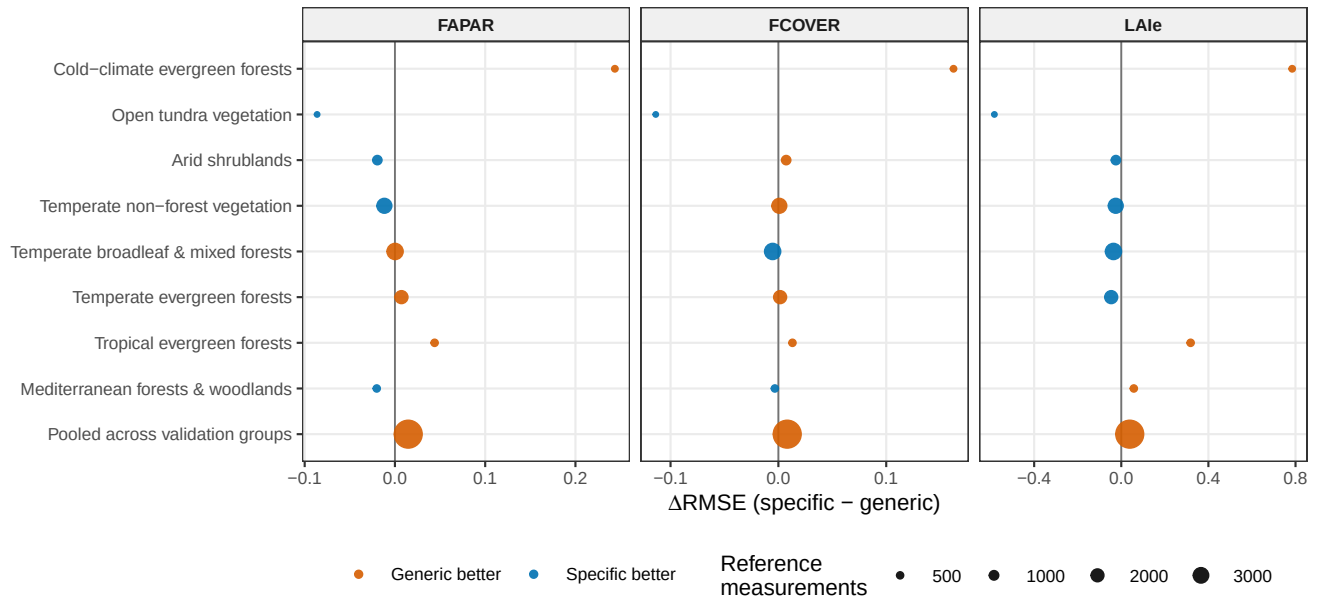


Figure S11: Change in out-of-sample RMSE for biome-vegetation-specific models relative to the generic model. Horizontal lines are paired individual-reference-measurement bootstrap 95% confidence intervals. Negative values indicate lower error for the specific model. Point size represents the number of reference measurements. These intervals do not account for dependence among measurements from the same ecoregion.

Table S10: Biome and NLCD land-cover combinations assigned to the eight validation groups. Biome names are followed by the included NLCD land-cover classes in italics; semicolons separate distinct mappings.

Validation group	Included biome–land-cover combinations
Cold-climate evergreen forests	Boreal Forests/Taiga, Tundra: <i>Evergreen Forest</i>
Open tundra vegetation	Tundra: <i>Grassland, Shrubland, Wetlands</i>
Arid shrublands	Deserts & Xeric Shrublands: <i>Shrubland</i>
Temperate non-forest vegetation	Temperate Broadleaf & Mixed Forests, Temperate Grasslands, Savannas & Shrublands: <i>Cropland, Grassland, Shrubland, Wetlands</i> ; Temperate Conifer Forests: <i>Grassland, Shrubland</i>
Temperate broadleaf & mixed forests	Temperate Broadleaf & Mixed Forests, Temperate Conifer Forests, Temperate Grasslands, Savannas & Shrublands: <i>Deciduous/Mixed Forest</i>
Temperate evergreen forests	Temperate Broadleaf & Mixed Forests, Temperate Conifer Forests, Temperate Grasslands, Savannas & Shrublands: <i>Evergreen Forest</i>
Tropical evergreen forests	Tropical & Subtropical Dry Broadleaf Forests, Tropical & Subtropical Moist Broadleaf Forests: <i>Evergreen Forest</i>
Mediterranean forests & woodlands	Mediterranean Forests, Woodlands & Scrub: <i>Deciduous/Mixed Forest, Evergreen Forest</i>

Table S11: Ecoregion-blocked out-of-sample performance of biome-vegetation-specific and generic models. R_g^2 uses the mean across all groups. ΔRMSE is specific minus generic; negative values indicate lower error for the specific model. Paired ecoregion-bootstrap 95% confidence intervals are reported for groups with at least 10 ecoregions; dashes indicate insufficient spatial replication.

Trait	Validation group	N	RMSE specific	RMSE generic	ΔRMSE	R^2 specific	R^2 generic	R_g^2 specific	R_g^2 generic
FAPAR	Cold-climate evergreen forests	377	0.467	0.224	+0.244	-1.930	0.329	-1.530	0.421
FAPAR	Open tundra vegetation	238	0.223	0.309	-0.086	-0.405	-1.705	0.510	0.057
FAPAR	Arid shrublands	949	0.046	0.066	-0.019	0.153	-0.705	0.988	0.977
FAPAR	Temperate non-forest vegetation	2762	0.122	0.133	-0.012	0.808	0.769	0.860	0.832
FAPAR	Temperate broadleaf mixed forests	3319	0.145	0.144	+0.000	0.687	0.688	0.822	0.823
FAPAR	Temperate evergreen forests	2034	0.212	0.205	+0.007	0.308	0.354	0.375	0.417
FAPAR	Tropical evergreen forests	513	0.229	0.185	+0.044	-0.310	0.145	-0.304	0.148
FAPAR	Mediterranean forests	504	0.183	0.203	-0.020	0.421	0.287	0.456	0.329
FAPAR	Pooled across validation groups	10696	0.179	0.164	+0.015	0.693	0.741	—	—
FCOVER	Cold-climate evergreen forests	377	0.389	0.226	+0.162	-1.199	0.255	-0.975	0.331
FCOVER	Open tundra vegetation	238	0.123	0.237	-0.114	0.636	-0.347	0.828	0.366
FCOVER	Arid shrublands	949	0.038	0.031	+0.007	0.385	0.596	0.991	0.994
FCOVER	Temperate non-forest vegetation	2762	0.114	0.113	+0.001	0.820	0.823	0.872	0.873
FCOVER	Temperate broadleaf mixed forests	3319	0.139	0.144	-0.005	0.737	0.716	0.844	0.832
FCOVER	Temperate evergreen forests	2034	0.187	0.185	+0.002	0.457	0.466	0.495	0.504
FCOVER	Tropical evergreen forests	513	0.207	0.194	+0.013	-0.099	0.035	-0.098	0.036
FCOVER	Mediterranean forests	504	0.179	0.182	-0.003	0.406	0.384	0.435	0.414
FCOVER	Pooled across validation groups	10696	0.159	0.151	+0.008	0.748	0.773	—	—
LAJe	Cold-climate evergreen forests	377	1.672	0.887	+0.785	-1.444	0.312	-1.390	0.327
LAJe	Open tundra vegetation	238	0.307	0.890	-0.582	0.250	-5.286	0.926	0.376
LAJe	Arid shrublands	949	0.073	0.098	-0.025	0.161	-0.502	0.997	0.994
LAJe	Temperate non-forest vegetation	2762	0.479	0.504	-0.025	0.676	0.641	0.810	0.789
LAJe	Temperate broadleaf mixed forests	3319	0.657	0.693	-0.036	0.727	0.696	0.835	0.816
LAJe	Temperate evergreen forests	2034	0.698	0.744	-0.046	0.469	0.396	0.477	0.405
LAJe	Tropical evergreen forests	513	1.041	0.722	+0.318	-0.620	0.220	-0.620	0.220
LAJe	Mediterranean forests	504	0.685	0.627	+0.058	0.247	0.369	0.424	0.517
LAJe	Pooled across validation groups	10696	0.679	0.640	+0.039	0.710	0.743	—	—